

KOMAL SABALE

+91 8421145187 || sabalekomal45@gmail.com || <https://www.linkedin.com/in/komal-sabale/> || <https://github.com/pleasecodekomal> || Pune, Maharashtra.

Education

Vivekanand Education Society's Institute of Technology, Mumbai

May 2026

BE Information Technology Engineering

CGPA: 8.65

Experience

Backend Developer Intern — QuizCred (Ongoing)

- Developed backend features for a **student assessment platform**, supporting student profiles, quiz registration, and performance tracking.
- Built APIs for **topic-wise performance analysis** and personalized learning recommendations.
- Supported the **admin dashboard backend** for quiz management, student analytics, and report exports.

Projects

Legal Chatbot

- Tools:** MERN Stack, FastAPI, Python ML Libraries
- Developed an AI-powered legal assistant to guide users through initial legal issue actions using conversational AI.
- Integrated NLP models including **Legal-BERT** (intent classification), **FLAN-T5** (response generation), and **MiniLM** (semantic search).
- Implemented MongoDB-based interaction logging to **support feedback-driven improvements** and model fine-tuning.
- Built a React.js chat interface with a FastAPI backend for scalable query processing and model inference.

RecruitML – Resume Screening and Recommendation System

- Tools:** Python, Scikit-learn, Logistic Regression, Natural Language Processing (NLP), HTML5/CSS3
- Engineered Hybrid AI Screening Pipeline:** Built a FastAPI & React screening engine combining SentenceTransformers (MiniLM) and skill-weight sliders, reducing manual resume evaluation time by 80%.
- Deployed ML Classifier & Explainable AI:** Trained a TF-IDF Scikit-learn model to generate natural language match verdicts (STRONG MATCH, NOT A MATCH) and automated career gap analyses.
- Architected Candidate RAG & Ethical Guardrails:** Integrated an interactive candidate Q&A RAG engine with single-toggle MD5 PII anonymization to ensure strict anti-bias hiring compliance.

FruitSense

- Tools:** Flutter, Python, Flask, TensorFlow/Keras, Scikit-learn, Google Gemini API
- Developed Cross-Platform AI Platform:** Built a Flutter and Flask web/mobile solution integrating CNN-based visual grading with supervised ML price forecasting across 41,200+ multi-fruit images (apples, bananas, oranges).
- Achieved High-Accuracy Quality Grading:** Trained fruit-specific CNNs achieving **91.6% overall test accuracy** on apples and **98.5% validation accuracy** on bananas, evaluating defects with an average response time of **3.5 seconds/image**.
- Integrated Quality-Aware Pricing & GenAI:** Combined CNN quality outputs with market parameters (variety, arrivals, region) to predict base prices at a **6.3% MAPE**, leveraging Google Gemini API to generate real-time qualitative price explanations.

Technical Skills

- Languages:** Python, C++, Java, JavaScript, SQL
- Web Development:** HTML, CSS, JavaScript, React.js, Node.js, FastAPI, MongoDB, Express.js, REST APIs
- Machine Learning:** Scikit-learn, Matplotlib, Transformers, TF-IDF, Cosine Similarity, NLP, Feature Engineering
- Database:** MySQL, PostgreSQL
- Tools & Platforms:** Git, GitHub, VS Code, Android Studio, PyCharm, Google Colab, Postman, Jupyter

Competitions

Nomura KakushIN 9.0 – Finalist | Project: ShoreCare: Website for NGO-Volunteer Cooperation in Beach Cleaning Drives.

- Contributed to **5+ core features**, improving operational efficiency and engagement for NGOs and volunteers
- Built the **CSR Partnership Toolkit** with automated impact dashboards and sponsor reports.
- Developed the **Beach Mapping** feature using a custom Beach Score (based on waste volume, tourist density, and local activities) to prioritize multiple cleanup sites.
- Designed **Volunteer Management** and **Skill-Based Role Tagging** systems, enhancing volunteer engagement and task distribution.